



Université de Sherbrooke
Summer school in biodiversity modelling August 25th, 2026

Realistic landscapes

Dominique Gravel

Exercise





30-30 Target

- ▶ Southern Québec has a deficit of protected areas
- ▶ Mostly small private areas
- ▶ Few large tracts of public lands

From metapopulation theory, should we aim at protecting as many private lands as possible, or should we put our efforts on public lands ?

Theory



Spatially explicit metapopulation dynamics

Each patch x is unique. We therefore need to track the probability it is occupied :

$$\frac{dp_x}{dt} = c_x P_x (1 - p_x) - e_x p_x$$

Note the difference in colonization, P_x is the total amount of propagules reaching patch x . We will define this quantity in a few slides.



What makes a patch unique ?



What makes a patch unique ?

- ▶ The quality for colonization
- ▶ The amount of propagules it produces
- ▶ It's spatial location
- ▶ Local persistence



What makes a patch unique ? Area

Population theory has established that extinction probability from demographic stochasticity $e \propto A^{-1}$

A first modification of the Levins model is therefore :

$$extinction = \frac{e_x}{A_x} p_x$$

where e_x could vary locally because of its intrinsic quality



What makes a patch unique ? Location

- ▶ Distance to neighbours
- ▶ State of neighbours
- ▶ Area of neighbours





What makes a patch unique ? Dispersal kernel

Propagules are more likely to reach nearby sites than distant ones. The minimal model for this hypothesis is the exponential function of distance :

$$K(x, y) = \exp(-\alpha d_{xy})$$

where α^{-1} is the average dispersal distance d_{xy}



What makes a patch unique ? Total dispersal

Now we need to aggregate dispersal from all neighbours. A first (wrong - but very common) approximation is :

$$P_x = \sum_{x \neq y}^N K(x, y) p_y A_y$$

1st note : it's the size of the patch producing propagules, not the one receiving propagules, that is size dependent.

2nd note : this formulation is not exact because it assumes that only one colonization occurs per unit time. It's however practical because it remains linear



Spatially explicit metapopulation dynamics [Full picture](#)

Plugging these definitions into the main equation we get :

$$\frac{dp_x}{dt} = c_x \sum_{x \neq y}^N K(x, y) p_y A_y (1 - p_x) - \frac{e_x}{A_x} p_x$$

For now we will consider parameters c and e to be constant across all locations (we'll get back to it at the end).



A compact notation

We can re-organize the equation using matrix notation.

Define :

$$m_{xy} = K(x, y)A_xA_y$$

where m_{xy} is the value in the entry $[x, y]$ of matrix M.

Matrix M therefore contains information on the spatial location of all patches, their contribution to colonization as well as the effect of area on extinction probability. All elements on the diagonal are set to 0.



For a simple solution

After a bit of magic (See Ovaskainen and Hanski 2002 for the maths), we find the very elegant condition for persistence :

$$\lambda_M > \frac{e}{c}$$

where λ_M is the first eigenvalue of the matrix M. A weighted average of the equilibrium vector of occupancies is :

$$p^* = 1 - \frac{1}{\lambda_M} \frac{e}{c}$$



Back to SLOSS

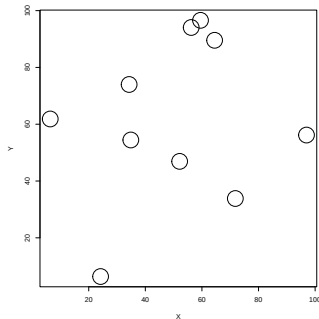
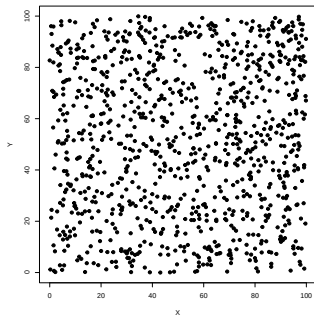
We now have a fairly simple tool to check the effect of patch size distribution on metapopulation persistence.

Two phenomenon happens at the same time for a constant total surface :

1. Larger patches have lower extinction probability and larger propagule production
2. Distance between patches increases as the number of patches decreases

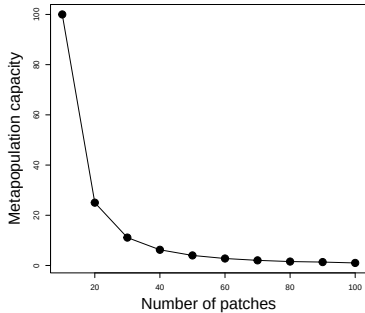


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Note : the basic model assumes a certain effect of patch area on metapopulation dynamics. It is possible to modify it with A^γ where γ is a tuning parameter.



Still an open debate

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EFFECTS OF HABITAT FRAGMENTATION ON BIODIVERSITY

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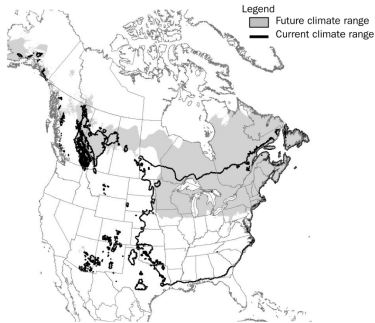
Key Words habitat loss, landscape scale, habitat configuration, patch size, patch isolation, extinction threshold, landscape complementation

■ Abstract The literature on effects of habitat fragmentation on biodiversity is huge. It is also very diverse, with different authors measuring fragmentation in different ways and, as a consequence, drawing different conclusions regarding both the magnitude and direction of its effects. Habitat fragmentation is usually defined as a landscape-scale process involving both habitat loss and the breaking apart of habitat. Results of empirical studies of habitat fragmentation are often difficult to interpret because (a) many researchers measure fragmentation at the patch scale, not the landscape scale and (b) most researchers measure fragmentation in ways that do not distinguish between habitat loss and habitat fragmentation per se, i.e., the breaking apart of habitat after controlling for habitat loss. Empirical studies to date suggest that habitat loss has large, consistently negative effects on biodiversity. Habitat fragmentation per se has much weaker effects on biodiversity that are at least as likely to be positive as negative. Therefore, to correctly interpret the influence of habitat fragmentation on biodiversity, the effects of these two components of fragmentation must be measured independently. More studies of the independent effects of habitat loss and fragmentation per se are needed to determine the factors that lead to positive versus negative effects of fragmentation per se. I suggest that the term “fragmentation” should be reserved for the breaking apart of habitat, independent of habitat loss.

And what about patch quality ?



Expected future distribution

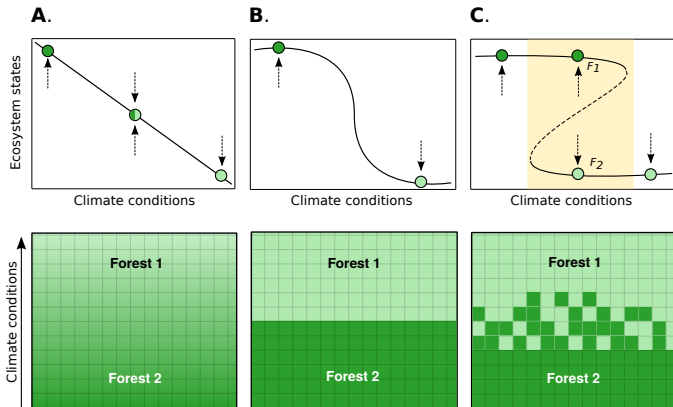


- ▶ Phenomenological;
- ▶ No demography;
- ▶ No biotic interactions;
- ▶ No dispersal limitations;
- ▶ Equilibrium distribution;

McKenney et al. 2007. Bioscience

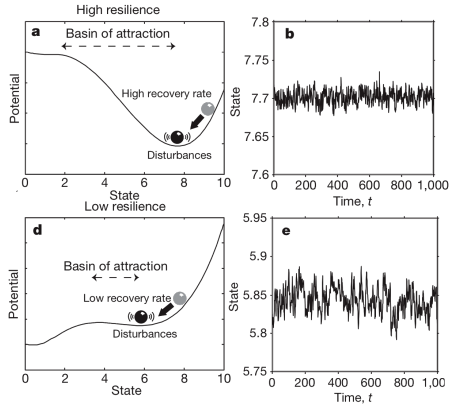


Non-linear transitions





Critical slowing down



Scheffer et al. (2009) Nature.



Model for range limits

Metapopulation dynamics:

$$\frac{dp(E)}{dt} = \underbrace{c(E)p(E)(h(E) - p(E))}_{\text{Colonisation}} - \underbrace{e(E)p(E)}_{\text{Extinction}}$$

Which yields the equilibrium distribution:

$$\bar{p}(E) = h(E) - \frac{e(E)}{c(E)}$$



Model for range limits Amount of change

The sensitivity of the system to a changing environment is thus given by:

$$\frac{\partial \bar{p}(E)}{\partial E}$$



Asymptotic rate of convergence

Consider a small deviation to the equilibrium under the new environmental conditions, E_1 , called δ :

$$\frac{d(\bar{p}(E_1) + \delta)}{dt} = \frac{d\bar{p}(E_1)}{dt} + \frac{d\delta}{dt}$$

With:

$$\delta = \frac{\partial \bar{p}(E_1)}{\partial E}$$

Linearization of the system around this equilibrium gives the dynamics of the small deviation (the leading eigenvalue in the case of a system of equations).



Asymptotic rate of convergence Metapopulation

Within the range limit:

$$\lambda = e(E) - c(E)$$

Outside the range limit:

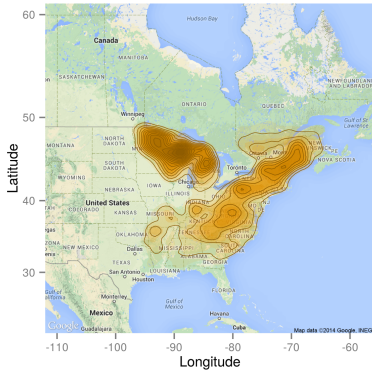
$$\lambda = c(E) - e(E)$$

At the range limit:

$$\lambda = c(E) - e(E) = 0$$



Data



- ▶ Permanent Forest Inventory plots from U.S. and Canada;
- ▶ Single tree measurements;
- ▶ 2-4 measurements;
- ▶ 5-15 years return interval;
- ▶ Classification in presence-absence;

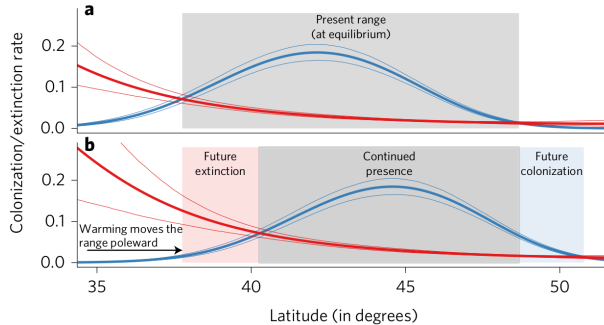


Fitting procedure

- ▶ Colonization and extinction models;
- ▶ Each transition is a binomial model;
- ▶ Approximation of the propagule pressure;
- ▶ Each transition is a function of annual average temperature and total precipitations;
- ▶ Bayesian parameter estimation with MCMC;



Solving for range limits

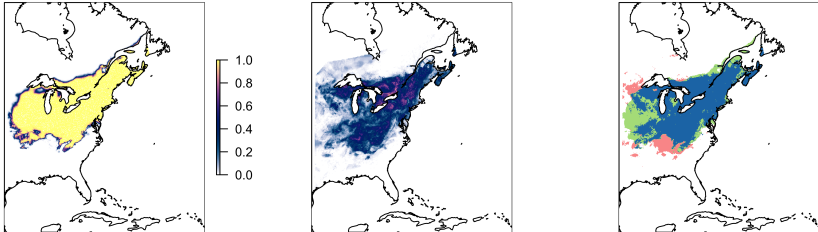


Talluto et al. 2017. Nature Ecology and Evolution



Solving for range limits

Acer saccharum

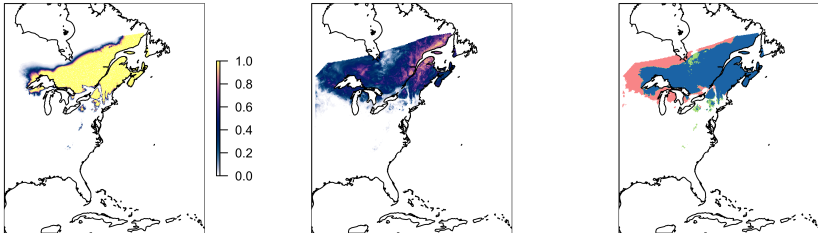


Talluto et al. 2017. Nature Ecology and Evolution



Solving for range limits

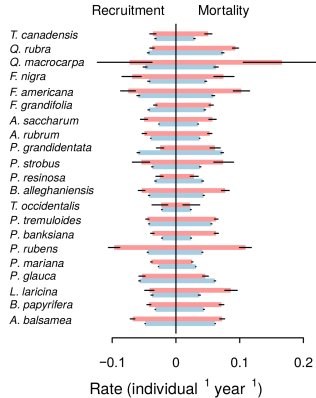
Abies balsamea



Talluto et al. 2017. Nature Ecology and Evolution



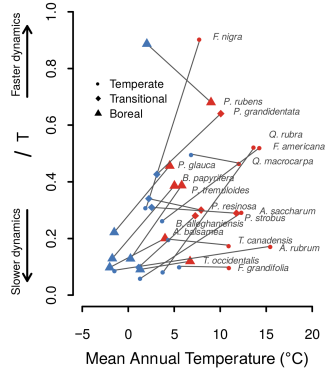
Demography



Talluto et al. 2017. Nature Ecology and Evolution



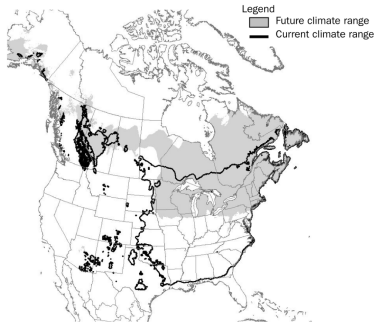
Rate of response



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Discussion Relevance of the theory

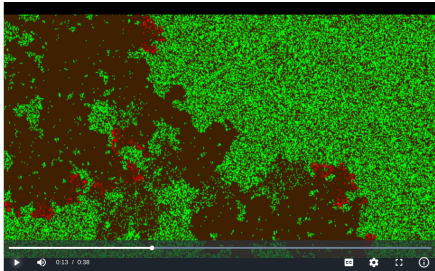


- ▶ A regional perspective on range limits appears to be more appropriate for forest ecosystems;
- ▶ The framework better accounts for transients;
- ▶ Many components of the response to climate change;
- ▶ The statistical approach solves several limitations of SDMs;

Final exercise : patch formation



Forest fire model Algorithm



- ▶ A burning cell turns into an empty cell
- ▶ A tree will burn if at least one neighbor is burning
- ▶ A tree ignites with probability f even if no neighbor is burning
- ▶ An empty space fills with a tree with probability p